

Artificial Intelligence

An Advanced Course About Foundations



Symbolic representation and inference: the intuitive idea

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An Aside: Syntax and Semantics

Symbolic Calculus

- *Solving* quadratic equations

$$x^2 + ax + b = 0$$

$$x^2 + 2(a/2)x + a^2/4 - a^2/4 + b = 0$$

$$(x + a/2)^2 - a^2/4 + b = 0$$

$$(x + a/2)^2 = a^2/4 - b$$

$$x = -a/2 \pm (a^2/4 - b)^{1/2}$$

A sequence of steps: at each step a *transformation rule* is applied

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From start to end

Start point: a premise (*we assume the truth of something*)

End point: a conclusion (*we state that something else must also be true*)

(*both points are selected by us*)

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From start to end

Start point: a premise (*we assume the truth of something*)

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(*both points are selected by us*)

Sequence of steps

Each transformation rule is based on a *semantic equivalence*

Each step must be *correct*, in the above sense

Symbolic Reasoning

Structures in Natural Language

- Natural language is not just a random collection of words:
it exhibits *systematic patterns* that allow generating and understanding
an infinite number of sentences from a finite set of elements

Sistematicity

No human being will understand the sentence:

*"Sally **likes** Harry"*

without being to understand:

*"Harry **likes** Sally"*

or any other sentences of the kind:

*"X **likes** Y"*

where X and Y can be any nouns or definite descriptions:

*"Ronald's girlfriend **likes** the cat that Linda brought home yesterday"*

(freely adapted from [Fodor e Phylyshyn, 1988])

Structures in Natural Language

- Natural language is not just a random collection of words: it exhibits *systematic patterns* that allow generating and understanding an infinite number of sentences from a finite set of elements

Compositionality

The meaning of a sentence is determined by the meanings of its parts and the rules used to combine them:

*"If **it rains**, (then) **the streets will be wet**"*

*"If **the streets will be wet**, (then) **we better drive carefully**"*

where the abstract pattern is:

*"If X **then** Y"*

and X, Y can be any clauses, even composite:

*"If **it rains** and **it's cold**, (then) **people stay indoors**"*

(freely adapted from [Fodor e Phylyshyn, 1988])

Reasoning and schemas: *syllogisms*

Many (intuitively) valid arguments obey an abstract schema

All (<i>humans</i>) are (<i>mortals</i>)	
All (<i>Greeks</i>) are (<i>humans</i>)	hence
All (<i>Greeks</i>) are (<i>mortals</i>)	

Abstract schema:

All H are M	
All G are H	hence
All G are M	

Reasoning and schemas: *syllogisms*

Many (intuitively) valid arguments obey an abstract schema

All (*humans*) are (*mortals*) hence

All (*Greeks*) are (*humans*)

All (*Greeks*) are (*mortals*)

Abstract schema:

All **H** are **M** hence

All **G** are **H**

All **G** are **M**

CAUTION!

*The position of the line is indeed relevant:
in this case the schema does not work*

Reasoning and schemas: *syllogisms*

Many (intuitively) valid arguments obey an abstract schema

All (<i>humans</i>) are (<i>mortals</i>)	
All (<i>Greeks</i>) are (<i>mortals</i>)	hence
All (<i>Greeks</i>) are (<i>humans</i>)	

Abstract schema:

All H are M	
All G are M	hence
All G are H	

CAUTION!

*The position with respect to the line is also relevant:
in this case the schema does not work*

Reasoning and schemas: *syllogisms*

Many (intuitively) valid arguments obey an abstract schema

All (<i>humans</i>) are (<i>mortals</i>)	
All (<i>Greeks</i>) are (<i>humans</i>)	hence
All (<i>Greeks</i>) are (<i>mortals</i>)	

Abstract schema:

All H are M	
All G are H	hence
All G are M	

The validity of schemas does not depend on *meaning*:

All (<i>enchanted frogs</i>) are (<i>princes</i>)	
All (<i>princes</i>) are (<i>young and beautiful</i>)	hence
All (<i>enchanted frogs</i>) are (<i>young and beautiful</i>)	

*Same schema,
different descriptions*

Fallacies (*paralogisms*)

Wrong sequence:

All (*humans*) are (*mortals*)

All (*Greeks*) are (*mortals*)

All (*Greeks*) are (*humans*)

*The last two sentences
have been switched*

Referential ambiguities:

(*Nothing*) is better than (*eternal happiness*)

(*One ham sandwich*) is better than (*nothing*)

(*One ham sandwich*) is better than (*eternal happiness*)

‘Obscure’ subtleties (obscure for now):

All (*enchanted frogs*) are (*princes*)

All (*enchanted frogs*) are (*young and beautiful*)

There is an (*enchanted frog*) which is (*a young and beautiful prince*)

What do we mean by “all”? Do we need the existence of at least one specimen to say “all”?

In such case, the schema would be valid

(*in formal logic it is not, as we will see*)

What is the purpose of *symbolic* logic?

*To distinguish correct reasoning
from incorrect reasoning*

What is the purpose of *symbolic* logic?

*To distinguish correct reasoning
from incorrect reasoning
by the **formal, symbolic** structure alone*

Truth and Possible Worlds: the quest for formal semantics

What is *true*?

- A world of *cats*

<i>likes</i>	Tom	Spot	Kitty	Felix
Tom	x			
Spot	x		x	
Kitty		x	x	
Felix			x	

- Sentences about this world

“Spot likes Tom” and “Tom does not like Spot”

“Tom likes himself”

“Kitty likes Spot” and “Spot likes Kitty”

“Kitty likes herself”

“Felix likes Kitty”

*All these sentences are **true**, in the world above*

What is *true*?

- A world of *cats*

<i>likes</i>	Tom	Spot	Kitty	Felix
Tom	x			
Spot	x		x	
Kitty		x	x	
Felix			x	

- Sentences about this world

“Spot likes Felix”

“Spot likes himself”

“Kitty does not like herself”

“Felix likes Spot”

*All these sentences are **false**, in the world above*

What is *true*?

- A world of *cats*

<i>likes</i>	Tom	Spot	Kitty	Felix
Tom	x			
Spot	x		x	
Kitty		x	x	
Felix			x	

- Sentences about this world (with *generalization*)

“Every cat likes a cat” is **true**

“Every cat likes another cat” is **false**

“Tom does not like any other cat” is **true**

“Kitty is liked by every cat” is **false**

“Every cat that likes another cat also likes her/himself” is ?

How to make a sentence *true*?

- Let's start from the other end: one sentence

"Kitty likes Spot" and "Spot likes Tom"

It can be made true in many different ways

Namely, it may be true in many different *possible worlds*

Examples (*possible worlds* of cats):

<i>likes</i>	Tom	Spot	Kitty	Felix
Tom				
Spot	x			
Kitty		x		
Felix				

<i>likes</i>	Tom	Spot	Kitty	Felix
Tom	x			
Spot	x		x	
Kitty		x	x	
Felix			x	

<i>likes</i>	Tom	Spot	Kitty	Felix
Tom	x	x	x	x
Spot	x	x	x	x
Kitty	x	x	x	x
Felix	x	x	x	x

How to make a sentence *true*?

- **Consider the sentence**

“Kitty likes Spot” and “Kitty does not like Spot”

There is no way to *make it true*

Namely, there is no *possible world* where this can be true

Sentences may be related

- Three sentences

- 1) "Every cat that likes Kitty likes Spot as well"
 - 2) "Tom likes Kitty"
-
- 3) "Tom likes Spot"

There is no way to make true sentences 1) and 2) without making sentence 3) true as well...
(just give it a try...)

This is just a small subset of all $2^{16} = 65,536$ possible worlds ...

likes	Tom	Spot	Kitty	Felix
Tom				
Spot	x			
Kitty		x		
Felix		x	x	

likes	Tom	Spot	Kitty	Felix
Tom				
Spot	x	x	x	
Kitty		x	x	
Felix		x	x	

likes	Tom	Spot	Kitty	Felix
Tom	x			
Spot			x	
Kitty		x		
Felix				x

likes	Tom	Spot	Kitty	Felix
Tom				
Spot	x	x	x	x
Kitty			x	
Felix		x	x	

likes	Tom	Spot	Kitty	Felix
Tom				
Spot				
Kitty			x	
Felix				

likes	Tom	Spot	Kitty	Felix
Tom	x	x	x	
Spot	x			
Kitty		x		
Felix		x	x	

likes	Tom	Spot	Kitty	Felix
Tom		x	x	
Spot	x	x	x	
Kitty	x	x		
Felix				

likes	Tom	Spot	Kitty	Felix
Tom	x	x	x	
Spot	x		x	
Kitty		x		
Felix		x		x

likes	Tom	Spot	Kitty	Felix
Tom	x	x	x	x
Spot		x		
Kitty			x	
Felix				x

likes	Tom	Spot	Kitty	Felix
Tom			x	
Spot	x			
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Felix		x	x	

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Sentence 1) is **true** in these worlds

likes	Tom	Spot	Kitty	Felix
Tom				
Spot	x			
Kitty		x		
Felix		x	x	

likes	Tom	Spot	Kitty	Felix
Tom				
Spot	x	x	x	
Kitty		x	x	
Felix		x	x	

likes	Tom	Spot	Kitty	Felix
Tom	x	x	x	
Spot	x			
Kitty		x		
Felix		x	x	

likes	Tom	Spot	Kitty	Felix
Tom		x	x	
Spot	x	x	x	
Kitty	x	x		
Felix				

likes	Tom	Spot	Kitty	Felix
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Tom	x	x	x	
Spot	x		x	
Kitty		x		
Felix		x		x

likes	Tom	Spot	Kitty	Felix
Tom	x	x	x	x
Spot		x		
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Spot				
Kitty			x	
Felix				

likes	Tom	Spot	Kitty	Felix
Tom	x	x	x	
Spot	x		x	
Kitty		x		
Felix		x		x

likes	Tom	Spot	Kitty	Felix
Tom			x	
Spot	x			
Kitty		x		
Felix		x	x	

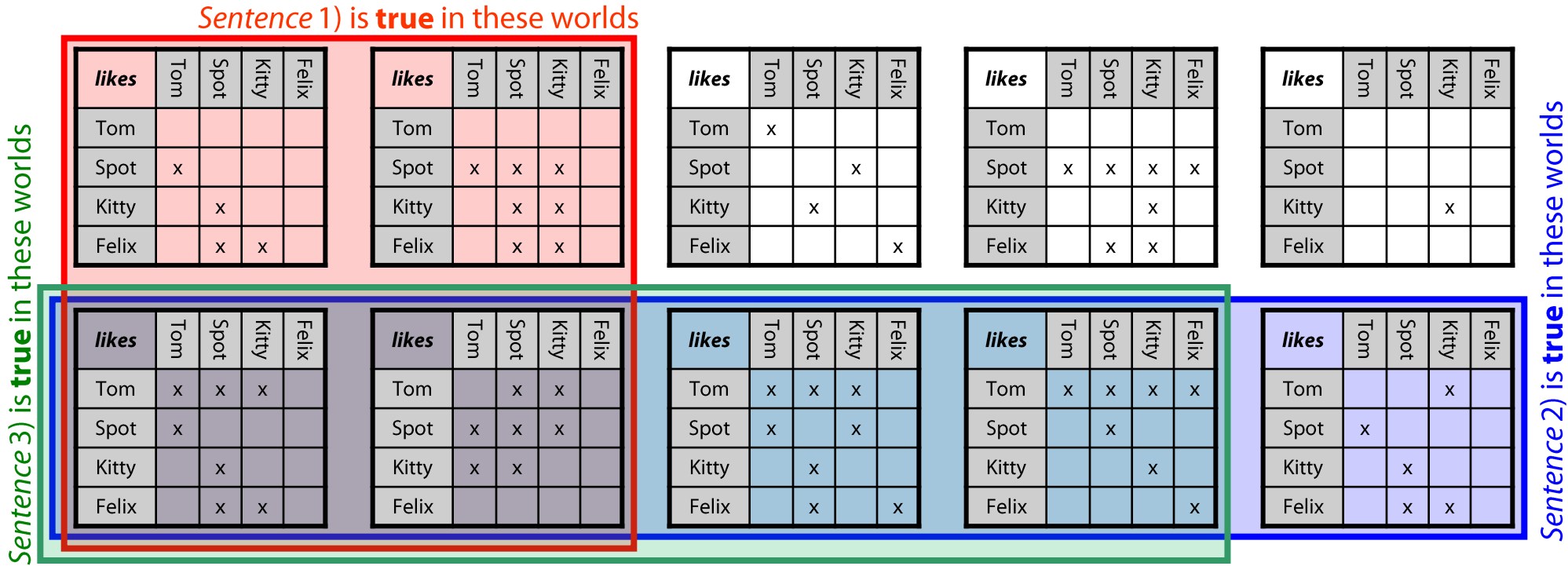
Sentence 2) is **true** in these worlds

Sentences may be related

■ Three sentences

- 1) "Every cat that likes Kitty likes Spot as well"
 - 2) "Tom likes Kitty"
-
- 3) "Tom likes Spot"

There is no way to make true sentences 1) and 2) without making sentence 3) true as well...
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Sentences may be related

- **Three sentences**

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(just give it a try)

Questions:

Was this just coincidence?

How many possible worlds must we consider, to be sure?

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Was this just coincidence?

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Hint:

All of them.